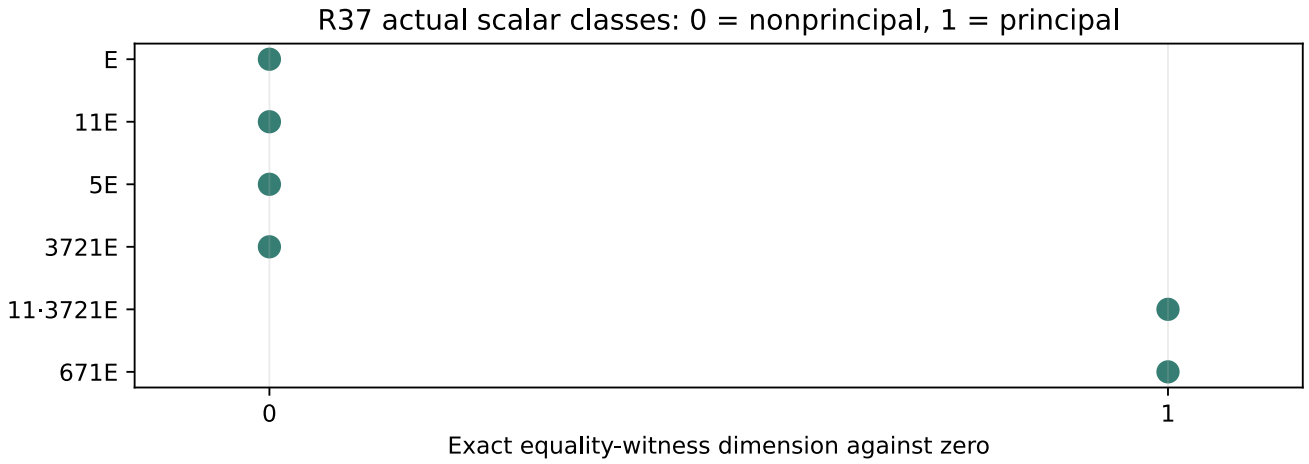
Koblitz transfers — R37

Exact source classes and finite cofactor; no IC performance claim

The actual deck order is bound through all four point coordinates, complete 83-power Frobenius orbits and680 saved literal function values. The signed class E is evaluated in two independent expressions, with equal results.



Both E and11E are nonprincipal. The official LMFDB coefficient data for the complementary form77.2.l.b give $P_b(-1)=3721=61^2$. All16 Hecke-basis inverse rows, the character/a4 recurrence and both independent matrix norms pass; the complete Euler polynomial of degree32 is retained.

The classes3721E and61(61E) agree and are nonprincipal, while11·3721E is principal. The predicted full mixed-carrier annihilator450241E is principal. The final671E test is principal. With the preceding nonzero multiples this gives exact source order671=11·61 and least positive cofactor61 to11-torsion, under the retained proper-curve/local/complete-RR/pole bridge. The actual saved class61E has order11. The finite projector122=2·61 is idempotent on this cyclic group, with value one modulo11 and zero modulo61.

All subspace divisions use every original generator, with residual/kernel, rank-nullity, recombination and final membership certificates. Every product is checked by two different arithmetic implementations. These internal checks are not an independent external scientific replay.

Lean: 9 audited declarations in 3 accepted runs; 2 rejected compile attempts are retained. The proofs check abstract weight, torsion, source-order and finite-projector implications. Native arrays, Picard implementation, normalization and factor maps are not formalized.

The intended CM-sign image remains unknown. Full quotient saturation, proper polarized embedding, theta, node187, arbitrary collectors and larger supports remain open. Native4GiB/1200s and pinned Lean-j1-M4096/1200s caps unchanged. Global metric integration is pending exclusive path ownership; scoped patch included. No timing comparison, IC relation rank or scalar recovery claim.

Sources: Khuri-Makdisi math/0409209v3; official LMFDB77.2.l / mf_hecke_nf; retained R15/R16 CM Euler/sign obstruction and R27 mixed-Prym bridge.